

T1057

Process Discovery

ATT&CK Detection Report • Home Lab

Date	2026-06-10
Technique	T1057 - Process Discovery
Test	#2 - Process Discovery via tasklist
Tactic	Discovery
Result	Detected

1. What This Technique Does

An attacker enumerates running processes on the victim machine to gather information about what software and security tools are active. This is reconnaissance, the attacker is building a picture of the environment before deciding their next move (such as weaponizing). Knowing what is running helps them avoid detection tools and identify high-value targets like lsass.exe for credential dumping.

2. Hypothesis

Expected process	tasklist.exe spawned by cmd.exe
Expected parent chain	powershell.exe → cmd.exe → tasklist.exe
Expected command line	cmd.exe /c tasklist
Expected event codes	EventCode=1 (process creation)

Expected Splunk search:

```
index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational"
host="DESKTOP-9KP1CU3" EventCode=1 CommandLine="*tasklist*
```

3. Execution

Command	Invoke-AtomicTest T1057 -TestNumbers 2
Exit code	0 (success)

Issues	None
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4. What Splunk Found

Image	C:\Windows\System32\tasklist.exe
CommandLine	tasklist
ParentImage	C:\Windows\System32\cmd.exe
ParentCommandLine	cmd.exe /c tasklist
User	DESKTOP-9KP1CU3\SOC101
Timestamp	2026-06-10 00:45:51
Event codes triggered	EventCode 1 (process create), EventCode 22 (DNS)

Working Splunk search:

```
index=main source="WinEventLog:Microsoft-Windows-Sysmon/Operational"
host="DESKTOP-9KP1CU3" EventCode=1
| where NOT match(Image, "(?i)splunk")
| table _time, Image, CommandLine, ParentImage, ParentCommandLine
| reverse
```

5. Hypothesis vs Reality

Got Right	Correctly predicted tasklist.exe, cmd.exe, and powershell.exe would be involved. Parent-child chain matched expectations.
Got Wrong	Expected a bigger or more dramatic attack footprint. Discovered that T1057 is a simple recon, it only generates process creation events because tasklist is a local command with no network or file activity.
Unexpected	Volume of background noise in Splunk. Filtering is a core skill, a MUST. Real SOC analysts spend significant time tuning searches to cut noise.

6. Detection Rule

Trigger Logic

ParentImage	*powershell.exe*
Image	*cmd.exe*
CommandLine	*tasklist*

Detection search:

```
index=main host="DESKTOP-9KP1CU3"  
source="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=1  
ParentImage="*powershell.exe*" Image="*cmd.exe*" CommandLine="*tasklist*"
```

False positive risk:

Low: legitimate software rarely chains

powershell → cmd → tasklist. Worth investigating every time it fires.

Alert Configuration

Name	alert_tasklist_attempt
Schedule	Every 5 minutes or */5 * * * *
Lookback window	Last 15 minutes
Trigger condition	Number of results > 0, per result
Severity	Medium
Mode	Per Result
Confirmed firing	2026-06-10 02:05:00

7. Cleanup & Next Steps

Cleanup command:

```
Invoke-AtomicTest T1057 -TestNumbers 2 -Cleanup
```

Next curiosity:

MITRE notes that process discovery can also use native API calls like `CreateToolhelp32Snapshot` instead of command line tools. This is stealthier because it leaves no `cmd.exe` or `tasklist.exe` process creation event. Want to explore what that looks like in Splunk and whether Sysmon can still catch it.